Zone Creation:

Zone Code (optional) Zone Type (mandatory)

[STR-B] [Storage ▼]

Zone Name (mandatory)

[Storage Zone B]

Max Weight (kg) Max Volume (cm^3)

[50000] [5000000]

Allowed Categories

[Electronics] [Hardware] [General] [Perishables] [Pharmaceuticals] [chemicals] [Fragile] [Heavy]

Handling Rules

[Add a handling rule...]

Do you want to add locations inside this zone now?

Number of Aisles (leave 0 if no aisles needed)

[0]

Number of Racks (leave 0 if no racks needed)

[0]

[Cancel] [Next →]

Two things to notice here. Admin fills zone details as usual. Then at the bottom two simple number fields appear — Number of Aisles and Number of Racks.

If admin types Aisles = 3 and Racks = 0 — system knows aisles exist and racks will be added inside aisles in the next step.

If admin types Aisles = 0 and Racks = 4 — system knows no aisles, racks go directly under zone.

If admin types Aisles = 0 and Racks = 0 — zone only, no locations now. Modal closes after saving.

Step 2A — If Aisles were entered (Aisles > 0)

System auto-generates aisle rows based on the number admin typed. Admin sees a table:

Configure your Aisles

Aisle Name	Aisle Code	Number of Racks
[Aisle 1]	[A1]	[2]
[Aisle 2]	[A2]	[3]
[Aisle 3]	[A3]	[2]

Names and codes are auto-filled but admin can edit them.

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Admin edits names if needed and types how many racks each aisle should have. Then clicks NEXT.

Step 2B — If No Aisles (Aisles = 0, Racks > 0)

Skip aisle step completely. Go directly to Rack step. System auto-generates rack rows based on number admin typed in Step 1:

Configure your Racks

Rack Name	Rack Code	Levels	Positions per Level
[Rack 1]	[R1]	[4]	[5]
[Rack 2]	[R2]	[4]	[5]
[Rack 3]	[R3]	[3]	[6]
[Rack 4]	[R4]	[0]	[8]

Note: If Levels = 0, positions are created directly under the rack.

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Step 3 — Rack Configuration (After Aisles Step)

If aisles exist — system now shows racks grouped under each aisle based on numbers admin entered in Step 2A:

Configure Racks

—— Aisle 1 (2 racks) ——

Rack Name Rack Code Levels Positions per Level

[Rack 1] [R1] [4] [5]

[Rack 2] [R2] [4] [5]

—— Aisle 2 (3 racks) ——

Rack Name Rack Code Levels Positions per Level

[Rack 1] [R1] [3] [4]

[Rack 2] [R2] [3] [4]

[Rack 3] [R3] [3] [4]

—— Aisle 3 (2 racks) ——

Rack Name Rack Code Levels Positions per Level

[Rack 1] [R1] [2] [6]

[Rack 2] [R2] [2] [6]

[← Back] [Next →]

Admin fills in levels and positions per level for each rack. If Levels = 0 for any rack, positions go directly under that rack.

Step 4 — Preview (Always shown before saving)

System calculates everything and shows a tree preview:

PREVIEW — What will be created:

Zone B (Storage) — STR-B

```
└── Aisle 1 — A1
    └── Rack 1 — R1
        └── Level 1 → P1, P2, P3, P4, P5
        └── Level 2 → P1, P2, P3, P4, P5
        └── Level 3 → P1, P2, P3, P4, P5
        └── Level 4 → P1, P2, P3, P4, P5
    └── Rack 2 — R2
        └── Level 1 → P1, P2, P3, P4, P5
        ...
└── Aisle 2 — A2
    ...
```

Total being created:

3 Aisles 7 Racks 24 Levels 120 Positions

[← Back] [Confirm & Save All]

The Entire Flow in One Simple Diagram

```
Step 1 — Zone Details
        + Number of Aisles
        + Number of Racks
            ↓
        ┌──────────────────────────────────┐
        │ Aisles > 0 ?                     │
        │ Yes → Step 2A (Aisle table)      │
        │ No  → Step 3 (Rack table)        │
        └──────────────────────────────────┘
            ↓
Step 2A — Aisle table
        (name, code, number of racks per aisle)
            ↓
Step 3  — Rack table
        (name, code, levels, positions per level)
        grouped under each aisle if aisles exist
            ↓
Step 4  — Preview tree
        Total count summary
            ↓
        Confirm & Save All
```

If Zone is Saved (With 0 Aisles and 0 Racks)

The modal closes. The new zone appears in the sidebar tree. But since it has no locations inside it, it shows an empty state with a clear prompt:

```
Zone B (Storage) — STR-B
  └── [ No locations yet — click + to add ]
```

When Admin Clicks the + Button on the Zone Node

A small clean popup/dropdown appears right next to that zone in the sidebar. Not a full modal — just a small contextual menu:

Add inside Zone B:

```
┌──────────────────────────────┐
│ + Add Aisle                  │
│ + Add Rack                   │
│ + Add Position directly      │
└──────────────────────────────┘
```

What Each Option Does

Add Aisle — opens a small form:

```
Aisle Name      Aisle Code      Number of Racks
[ Aisle 1 ]     [ A1 ]          [ 2 ]
                        [ + Add Another Aisle ]

[ Cancel ]    [ Next → ]
```

After admin fills aisles and clicks Next — goes straight to Rack configuration table we discussed. Same flow as during zone creation.

Add Rack — opens rack form directly under this zone (no aisle):

```
Rack Name    Rack Code    Levels    Positions per Level
[ Rack 1 ]   [ R1 ]       [ 4 ]     [ 5 ]
          [ + Add Another Rack ]

[ Cancel ]    [ Next → ]
```

After Next — shows preview then Save.

Add Position directly — for very simple warehouses that just need zone and positions with no rack or aisle at all:

Number of Positions: [10]
Starting Code: [P001]

[Cancel] [Save]

Same Thing for Aisle, Rack, and Level Nodes

When admin clicks + on an Aisle node:

Add inside Aisle 1:

```
+ Add Rack
```

Only one option because aisle can only contain racks.

When admin clicks + on a Rack node:

Add inside Rack 1:

```
+ Add Level
+ Add Position directly
```

Add Level — opens:

Number of Levels: [4]
Positions per Level: [5]

[Cancel] [Save]

Add Position directly — opens:

Number of Positions: [10]
Starting Code: [P001]

[Cancel] [Save]

When admin clicks + on a Level node:

Add inside Level 1:

```
+ Add Position
```

Only one option because level can only contain positions.

Complete Picture — Both Ways Work Together
WAY 1 — During Zone Creation (All at once)
Zone form → type numbers → auto-generate
→ fill rack table → preview → Save All

WAY 2 — After Zone is Saved (Add later)
Click + on any node in sidebar
→ contextual menu appears
→ choose what to add
→ small focused form opens
→ save

Same result. Same data. Admin chooses whichever way is more convenient for them at that moment. Both ways are always available.

[Update canvas rendering to draw the full hierarchy visually. Zone = large colored box. Aisle = labeled corridor strip inside zone. Rack = box inside aisle or zone. Level = horizontal shelf row inside rack. Position = individual colored cell inside level row

Position cell colors stay the same — empty white, partial orange, full green, reserved blue, locked red]

This change affects every module that shows bin location:

- Putaway module
- Pick Lists
- Stock Levels tab
- Movements tab
- Cycle Count
- Stock Transfer
- GRN putaway suggestion
- Reports